Operating Systems

Assignment 1: Part A – Theory

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TQ1

1. 120_{ms}
2. 40_{ms}
3. 4.8_{ms}

TQ2

1. 1,2
2. 1: Context switch after line 1 or line 2 the value stays at 1. 2: The context switch happens after either A or B run the code (no interruption).

TQ3

1. Ticket lock guarantees FIFO acquisition order. Let $x, y \in \mathbb{N}\ such\ that\ x < y$. Suppose Thread 1 has ticket x and Thread 2 has ticket y. Suppose the ticket counter $= x$. Since the ticket counter is only incremented after the critical section of the thread that is acquiring the ticket, Thread 2 cannot acquire ticket y until Thread 1 has acquired ticket x as $x < y$. Therefore, when the ticket counter increases, Thread 1 will recognize that their ticket is available before the ticket counter reaches y.
2. Ticket lock guarantees bounded waiting. Let $k, c \in \mathbb{N}\ such\ that\ c < k$ and suppose that Thread 1 gets ticket k. Since the ticket lock is FIFO, there is at most $k - c$ threads that can enter before it Thread 1 and therefore the ticket lock guarantees bounded waiting.
3. Suppose that Thread 1 is waiting for their turn. Since there is no FIFO or designated order for a thread to get their turn, Thread 1 can wait forever as two or any number of other threads can keep taking their turn while Thread 1 never gets it and therefore is starved.

TQ4

1. Suppose that the semaphore is empty and the Consumer acquires the mutex first. The consumer will then wait until the semaphore is full forever. Even with a context switch, because the mutex is mutually exclusive, the Producer cannot acquire the mutex to insert an item.

2. We swap lines 1 and 2 for both the producer and the consumer so that we never acquire the mutex unless we can modify (and pass the wait).